Secure File Storage System Using AES Encryption

Introduction

With increasing digital data, securing files has become essential. This project focuses on protecting files using encryption techniques.

Abstract

The Secure File Storage System is developed using Python to encrypt and decrypt files. It uses AES-based encryption through the Fernet module. Only authorized users with the correct key can access the original file, ensuring data security.

Tools Used

- Python
- Cryptography (Fernet)
- VS Code

Steps Involved

1. Generate and store an encryption key.
2. Read file data in binary format.
3. Encrypt the file using the key.
4. Save encrypted file with .enc extension.
5. Decrypt the file using the same key.
6. Restore original content securely.

Conclusion

The project shows how encryption can secure files from unauthorized access. It is a simple yet effective implementation of data protection techniques.